From Generation to Adaptation: Gradual Diffusion Transitions for Cross-Domain Hyperspectral Image Classification

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***Abstract*—Cross-domain hyperspectral image (HSI) classification via unsupervised domain adaptation (UDA) is practical for Earth observation, yet its performance is often limited by severe distribution shifts. Existing methods typically seek a one-step nonlinear mapping in a highly nonconvex optimization space, which risks representation collapse and semantic distortion. To address this issue, we propose a diffusion-guided manifold evolution adaptation (Diff-MEA) framework. Instead of using generative models merely for data augmentation, the Diff-MEA reformulates the UDA as a continuous manifold evolution process that decouples macroscopic domain gaps into tractable microshifts via gradual diffusion transitions. To facilitate this evolution, we introduce a synergistic semantic-boundary coevolution (SSBCE) mechanism, which alternates between a semantic anchoring (SA) subphase to penalize generative drift and a boundary adaptation (BA) subphase to accommodate target-shifted manifolds. To realize these phases, a memory-transformer domain adapter (MTDA) integrates salient peak response selection (SPRS) and local feature mean aggregation (LFMA) to construct persistent memory tokens, which serve as robust semantic anchors and support dynamic decision boundary expansion. Extensive experiments on three benchmark datasets demonstrate that the Diff-MEA consistently outperforms the state-of-the-art methods in handling severe domain gaps. The code is available at https://github.com/B-Xi**

Received 15 May 2026; revised 26 July 2026; accepted 2 September 2026. Date of publication 14 September 2026; date of current version 2 October 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62401434 and Grant 62671482, in part by the National Natural Science Foundation for Distinguished Young Scholar under Grant 62425103, in part by the Fundamental Research Funds for the Central Universities under Grant QTZX26121, in part by the Young Talent Support Program of Xi'an City Association for Science and Technology under Grant 0959202513120, in part by China Postdoctoral Science Foundation under Grant 2023M732742, and in part by the Discipline Innovation and Talent Introduction Base of Colleges and Universities in Shaanxi Province. *(Corresponding authors: Yunsong Li; Zan Li.)*

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Digital Object Identifier 10.1109/TGRS.2026.3733406

***Index Terms*—Diffusion model, hyperspectral image (HSI) classification, progressive alignment, unsupervised domain adaptation (UDA).**

I. Introduction

DRIVEN by advances in sensor technology and algorithms, hyperspectral images (HSIs) can capture intrinsic spectral signatures across hundreds of narrow bands [1], [2]. By integrating rich spectral and spatial information, the HSI enables fine-grained pixel-wise classification [3], [4]. This capability underpins a wide range of applications, including agricultural monitoring and environmental assessment [5], [6].

Motivated by the need for precise classification, HSI analysis has evolved from traditional machine learning with handcrafted features to sophisticated deep learning paradigms [7], [8]. By automatically learning hierarchical representations, architectures such as multidimensional convolutional neural networks (1-D, 2-D, and 3-D) [9], [10], [11], and self-attention-based transformers [12] have achieved remarkable success in modeling complex spectral–spatial correlations. In contrast to the computationally intensive self-attention mechanism in transformers, structured state space models (SSMs) have recently emerged as efficient alternatives. In particular, a lightweight architecture known as Mamba [13] stands out for long-range dependency modeling. Fundamentally, Mamba integrates a data-dependent selective scan mechanism into its recurrent dynamics. This architectural innovation empowers the model to dynamically filter out irrelevant noisy bands while selectively remembering critical context across extended sequences [14]. For HSI analysis, this mechanism provides a distinct advantage that enables the precise modeling of continuous, long-range spectral dependencies with linear time complexity [15]. However, these data-driven approaches are fundamentally designed for single-domain scenarios and lack effective knowledge transfer mechanisms. Consequently, despite strong performance within the same domain, they remain highly sensitive to spectral distribution shifts, suffering precipitous performance degradation when deployed in cross-scene environments [16].

To address this challenge, recent studies have increasingly turned to domain adaptation, which aligns feature distributions across domains while preserving task-relevant

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semantics, thereby enabling the learning of domain-invariant representations [17], [18]. Among these paradigms, unsupervised domain adaptation (UDA) that leverages labeled source data and unlabeled target data has emerged as a dominant strategy for knowledge transfer [19], [20], [21]. Early UDA methods primarily include instance-based, classifier-based, and feature-based approaches, most of which rely on shallow, handcrafted features. In particular, instance-based techniques reweight source samples to approximate the target distribution. Classifier-based methods aim to construct domain-robust decision boundaries, whereas feature-based strategies minimize statistical discrepancies within a shared latent space. Building upon these foundations, deep learning-based UDA methods have evolved significantly in recent years. In particular, distance-based UDA techniques seek to extract domain-invariant features by minimizing the statistical divergences between source and target feature distributions [22], [23].

Another prominent line of work explores adversarial training, where a domain discriminator is trained against a feature extractor in a min–max game to learn domain-agnostic representations. Ganin and Lempitsky [24] pioneered this direction with the domain-adversarial neural network (DANN), which encourages the feature extractor to produce domain-confused representations under unsupervised settings. More recently, Zhang et al. [25] proposed TSTnet, a topology- and semantic-aware transfer network that models global object relationships via graph convolutional networks (GCNs) and aligns feature distributions using graph optimal transport to preserve topological consistency. Zhu et al. [26] further refined global alignment through the deep subdomain adaptation network (DSAN), which aligns class-conditional subdomain distributions using local maximum mean discrepancy (LMMD), thereby enabling more fine-grained adaptation.

Beyond traditional alignment strategies, recent efforts have incorporated auxiliary mechanisms to refine the distribution matching. For instance, Li et al. [27] introduced supervised contrastive learning for UDA (SCLUDA) to enhance feature discriminability on labeled source data prior to adversarial adaptation. To better bridge domain gaps, Cai et al. [28] proposed a multilevel UDA (MLUDA) scheme that aligns representations across the image, feature, and logit spaces. Furthermore, Fang et al. [29] developed a masked self-distillation domain adaptation (MSDA) framework that combines class-specific adversarial training with pseudolabel guidance from a teacher model to mitigate cross-scene discrepancies. To address pseudolabel noise and complex domain variations, Liang et al. [30] designed a spectral structure-aware initialization and probability-consistent self-training (S2PST) approach. This method utilizes nuclear-norm maximization to calibrate target probabilities and prevent source overfitting. From a feature decomposition perspective, Feng et al. [31] introduced the shift-sensitive spatial–spectral disentangling learning (S^4DL), which leverages gradient-guided masks to strip domain-specific redundancy and dynamically adjusts the disentanglement strength based on cross-domain variance.

Despite these advancements, prevailing paradigms exhibit profound limitations under substantial domain shifts, primarily due to their reliance on single-step hard alignment [32]. Conventional approaches aim to directly optimize a global mapping function $\phi : \mathcal{X} \rightarrow \mathcal{Z}$ that aligns the source distribution P_s and target distribution P_t within a highly nonconvex manifold [33]. When the domain discrepancy involves severe geometric misalignment, forcibly minimizing the distributional divergence often precipitates representation collapse and reduced class discriminability. In high-dimensional spaces, P_s and P_t typically reside on distinct manifolds separated by substantial geodesic distances. Attempting to bridge this gap via aggressive, single-step alignment frequently induces feature topology collapse. In particular, aggressive single-step alignment may therefore distort the intrinsic feature topology by suppressing class-discriminative structures in pursuit of global distribution matching.

While denoising diffusion probabilistic models (DDPMs) have been predominantly utilized for data generation in HSI [34], their intrinsic Markov chain architecture inherently facilitates gradual, controllable cross-domain mapping. This property motivates our proposed DDPM-based UDA paradigm. Mathematically, varying diffusion timesteps induce a sequence of intermediate probabilistic states that progressively bridge the source and target manifolds. Through incremental Gaussian perturbations, the diffusion process decomposes abrupt source-to-target alignment into a series of gradual transitions, thereby reducing the risk of feature topology collapse.

Motivated by these theoretical insights, we propose the diffusion-guided manifold evolution adaptation (Diff-MEA) framework for cross-domain HSI classification. Different from conventional paradigms that rely on single-step statistical moment matching or structural contrastive alignment, the Diff-MEA reformulates the UDA as a continuous probabilistic transport process. By harnessing the forward and reverse Markov chains inherent in diffusion models, our framework decomposes the severe domain discrepancies into a sequence of tractable, intermediate manifold transitions. Furthermore, to ensure this probabilistic mapping strictly preserves class-discriminative topologies, we seamlessly integrate generative distribution evolution with discriminative semantic enhancement. This novel formulation enables source representations to be gradually aligned with the target manifold, thereby mitigating negative transfer.

The main contributions of this work are as follows.

1) A novel generative framework, Diff-MEA, is proposed, which formulates the UDA as a continuous, diffusion-driven manifold evolution process. By replacing aggressive single-step hard alignment with a multistep probabilistic transport mechanism, this approach effectively alleviates feature distortion and semantic erosion under severe domain shifts.
2) To explicitly model the intermediate domain representations, an iterative optimization strategy featuring synergistic semantic-boundary coevolution (SSBCE) is designed. By alternating between semantic anchoring (SA) and boundary adaptation (BA) subphases, this strategy constructs a smooth, traversable path, ensuring progressive adaptation across significant domain gaps.

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3) We introduce the memory-transformer domain adapter (MTDA), which integrates salient peak responses and local features into memory tokens to preserve rich regional contextual information. Functioning as a dynamic engine within the SSBCE loop, the MTDA provides robust semantic anchors to regularize the generative diffusion process, thereby enabling adaptive decision boundary refinement under domain shifts.

The remainder of this article is organized as follows. Section II reviews related works on diffusion models and UDA. Section III details the proposed Diff-MEA framework. Section IV presents extensive experiments and ablation studies on the benchmark datasets. Finally, Section V concludes the study.

II. Related Works

A. Diffusion Models

Diffusion models are latent-variable generative frameworks grounded in nonequilibrium thermodynamics. In particular, they operate through a forward Markov chain that progressively corrupts data distributions by injecting Gaussian noise, and a corresponding reverse process that is typically learned via variational inference to reconstruct clean samples. Following their early success in natural language processing and molecular generation [35], [36], diffusion models have become increasingly prevalent across various domains.

Currently, the theoretical landscape of diffusion models generally converges into three main paradigms: score-based generative modeling, continuous-time stochastic differential equations (SDEs), and the canonical DDPM framework [37]. Beyond their generative capabilities, diffusion models have demonstrated notable robustness and flexibility in various vision tasks, including image inpainting, translation, and fusion [38]. Furthermore, their learned latent representations can be effectively transferred to discriminative tasks such as classification, detection, and segmentation.

Motivated by these robust modeling capabilities, recent research has increasingly explored diffusion priors for cross-domain adaptation tasks. For example, Yu et al. [39] proposed the intercoder connected latent diffusion (ICCLD) method, which employs adversarial learning within a latent space to align source and target distributions. More recently, He et al. [40] introduced the diffusion domain teacher (DDT) framework, utilizing a frozen, pretrained diffusion model to synthesize high-fidelity pseudolabels for unlabeled target data, thereby supervising a student detector.

B. Unsupervised Domain Adaptation

UDA addresses the fundamental challenge of transferring task-relevant knowledge from a labeled source domain to an unlabeled target domain under distribution shifts. Over the past few years, deep UDA methodologies have evolved into several dominant paradigms to bridge this gap. Adversarial adaptation frameworks, for instance, align feature spaces by formulating a minimax game to disentangle domain-invariant and domain-specific representations [19], [41]. By pitting a feature extractor against a domain discriminator, these methods strive to reach an equilibrium where target features become statistically indistinguishable from source features. Concurrently, discrepancy-based approaches focus on explicitly minimizing statistical divergences, such as optimal transport or maximum mean discrepancy, across distinct domains [42]. This paradigm typically projects data into a shared latent space, strictly regularizing the statistical moments to ensure global distribution matching. To further refine decision boundaries and alleviate target-domain ambiguity, self-training and pseudolabeling techniques have been widely adopted [43]. These strategies iteratively propagate confident predictions to target samples, often employing advanced filtering mechanisms to mitigate the detrimental effects of noisy labels and confirmation bias.

In terms of HSI, the intrinsic high-dimensionality and complex spatial–spectral structures demand tailored architectural designs. Consequently, recent efforts have integrated advanced architectures, such as transformers and graph convolution networks, to impose robust spatial–spectral structural constraints and capture long-range dependencies during the cross-scene adaptation process [44], [45]. However, these prevailing techniques predominantly rely on single-step hard alignments, rendering them inherently vulnerable to feature topology collapse. To address this issue, the proposed Diff-MEA circumvents the drawbacks of abrupt alignment by reformulating UDA as a continuous, multistep probabilistic transport process, thereby facilitating stable and topology-preserving manifold evolution.

III. Proposed Diff-MEA for UDA

As illustrated in Fig. 1, the proposed Diff-MEA framework is organized as a four-phase pipeline that progressively enables cross-domain adaptation through structured feature evolution. In phase I, a Mamba-based encoder is pretrained solely on SD and subsequently frozen to provide stable latent representations. Based on these embeddings, the MTDA module applies salient peak response selection (SPRS) and local feature mean aggregation (LFMA) to capture salient spectral peaks and regional contextual statistics. The resulting memory tokens are concatenated with the original features and processed through parallel spatial–spectral branches, preserving spectral identities while accommodating spatial variations. In phase II, a DDPM is trained on TD latent features Z_t to learn the marginal data distribution, thereby establishing a target-aware generative prior. Phase III forms the core optimization stage governed by the SSBCE mechanism, which alternates between the SA and the BA subphases. During the SA subphase, progressively perturbed SD features Z_s^k are denoised by the DDPM to produce transitional representations $\tilde{Z}^k$, while only DDPM parameters are updated using $\mathcal{L}_{sa}$ to prevent semantic distortion. In the BA subphase, the DDPM is fixed, and the MTDA module is optimized via $\mathcal{L}_{ba}$, enabling gradual assimilation of target-shifted manifolds and controlled expansion of decision boundaries. As the diffusion timestep (k) increases across training epochs, the transitional domain progressively approaches the TD distribution. Finally, phase IV integrates the frozen Mamba-based encoder with the optimized MTDA to perform reliable cross-scene inference on the TD.

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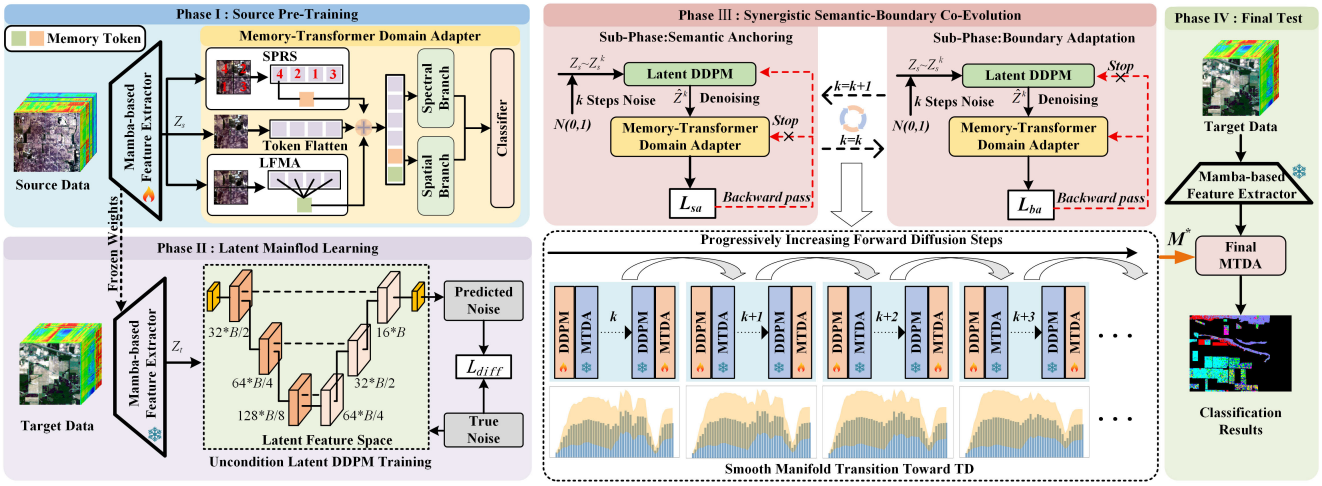


Fig. 1. Overall architecture of the proposed Diff-MEA for cross-scene HSI classification.

A. Pretraining in Phase I

To effectively capture the high-dimensional feature representations of HSI, the backbone network is pretrained on the SD during phase I. This backbone comprises two primary components: a Mamba-based encoder and the MTDA module. In particular, the encoder employs a spatial–spectral scanning mechanism as the Mamba architecture [15]. The MTDA module and the pretraining process are detailed as follows.

1) Memory-Transformer Domain Adapter: As shown in Fig. 2, the MTDA module first processes the input feature maps using a $1 \times 3 \times 3$ 3D convolutional layer, followed by a 3×3 2D convolutional layer. To isolate peak response variables along the feature manifold Ω, the SPRS employs a global spatial maximum operation. By capturing the maximum feature response, the SPRS extracts salient, category-specific cues that are critical for preserving class boundaries throughout the generative process. Formally, the SPRS is defined as follows:

$$\Phi_{SPRS}(TR_c) = \max_{p \in \Omega} TR_c(p) \tag{1}$$

where $TR_c(p)$ is the feature response of the cth channel, p denotes the spatial position, Ω represents the spatial domain of the feature map, and $|\Omega|$ indicates the total number of spatial pixels. Concurrently, the LFMA serves as a probabilistic spatial expectation estimator. To capture the holistic spectral–spatial context, it computes the first-order statistical moment over the same spatial domain

$$\Phi_{LFMA}(TR_c) = \mathbb{E}_{p \in \Omega}\left[TR_c(p)\right] = \frac{1}{|\Omega|}\sum_{p \in \Omega} TR_c(p). \tag{2}$$

These distinct spatial priors are subsequently concatenated to formulate the memory sequence

$$Token_s = [\Phi_{LFMA}(TR_c) \,\|\, \Phi_{SPRS}(TR_c)] \tag{3}$$

where $\|$ denotes the concatenation operation. This procedure yields a robust set of 72 memory tokens that encapsulate comprehensive structural details and mitigate information loss. Subsequently, let the input sequence be defined as follows:

$$X = \left\{Token_0^{cls}, x_1, \ldots, x_n\right\} + PE_{pos} \tag{4}$$

where $Token_0^{cls}$ represents the learnable class token and PE_{pos} denotes the positional embeddings. The sequence X and the memory sequence Token_s are transformed into query, key, value, and memory embeddings via linear projections

$$Q = XW_q, \quad K = XW_k, \quad V = XW_v, \quad S = Token_s W_s \tag{5}$$

where W_q, W_k, W_v, and W_s are trainable weight matrices. To inject global contextual priors into the attention mechanism, these projections are partitioned into h heads, and the memory tokens are concatenated with the keys and values

$$\tilde{K}^{(j)} = \left[K^{(j)} \| S^{(j)}\right], \quad \tilde{V}^{(j)} = \left[V^{(j)} \| S^{(j)}\right]. \tag{6}$$

The attention output for the jth head is then computed as follows:

$$O^{(j)} = \text{softmax}\left(\frac{Q^{(j)}\left(\tilde{K}^{(j)}\right)^{\top}}{\sqrt{\gamma}}\right)\tilde{V}^{(j)} \tag{7}$$

where γ is a scaling factor corresponding to the head dimension. Following the multihead attention (Attn) and multilayer perceptron (MLP) operations, the representations are updated via residual connections to facilitate optimization across deeper layers:

$$Y_\ell = X_\ell + \text{Attn}(\text{LN}(X_\ell)) \tag{8}$$

$$X_{\ell+1} = Y_\ell + \text{MLP}(\text{LN}(Y_\ell)) \tag{9}$$

where LN represents layer normalization and $\ell \in [1, L]$ indicates the index of the iteration transformer block. Within the lightweight dual-branch architecture, the spectral branch employs a 1×1 pointwise 2-D convolutional layer combined with multihead attention to selectively blend spectral channels. In parallel, the spatial branch utilizes a 3×3 2-D convolutional layer coupled with multihead attention to extract fine-grained local textures. Finally, the outputs from both branches are fused via a 1×1 2-D convolutional layer and a linear projection.

Moreover, this memory-augmented dual-branch architecture is essential for handling highly dynamic intermediate features during iterative coevolution. By integrating prominent semantic peaks with smooth contextual backgrounds, the evolving

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representations are projected into dynamically updated memory tokens. This dual-perspective modeling enhances feature resilience across continuous domain transitions. As the DDPM progressively deforms the manifold, the MTDA consistently captures underlying spatial geometries. Consequently, rather than experiencing the semantic drift often observed in standard models, these memory tokens serve as reliable structural anchors that facilitate stable and precise cross-domain adaptation.

In particular, during the SA subphase, memory tokens act as domain-invariant prototypes, injecting stable categorical priors to anchor semantic identities against topological collapse while the DDPM alters the data distribution. In the subsequent BA subphase, the decoupled dual-branch structure dynamically leverages spatial context to compensate for spectral distortions caused by domain shifts. This synergy ensures precise refinement of decision boundaries across nonconvex manifolds.

2) Pretraining: To establish a reliable initial representation, the model is first pretrained on the labeled SD. Let $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ denote the source dataset, where x_i^s is the ith input sample, N_s is the total number of samples, and $y_i^s \in \{1, \ldots, N_c\}$ is the corresponding ground-truth label among N_c categories. During this phase, the Mamba-based encoder f_ψ projects the inputs into a latent space, and the MTDA $\boldsymbol{M}$ outputs the predicted class probabilities. The network is optimized using the standard cross-entropy loss $\mathcal{L}_{\text{cls}}$, formulated as follows:

$$\mathcal{L}_{cls} = -\mathbb{E}_{(x_i^s, y_i^s)\sim\mathcal{D}_s} \log\left[\boldsymbol{M}\left(f_\psi\left(x_i^s\right)\right)\right] \tag{10}$$

where $[\cdot]_{y_i^s}$ explicitly extracts the predicted probability corresponding to the true class label y_i^s.

B. DDPM Training in Phase II

In Phase I, the pretrained Mamba-based encoder f is derived for robust feature extraction. This frozen module then projects the target samples $\mathcal{D}_t = \{x_j^t\}_{j=1}^{N_t}$ into a deterministic latent set $Z_t = \{z_j^t\}_{j=1}^{N_t}$, where $z_j^t = f(x_j^t)$. These latent representations serve as the foundational training data for the subsequent feature-space diffusion. Operating in the latent space mitigates computational overhead by compressing high-dimensional pixel data into compact, semantically rich tokens while preserving the intrinsic spectral–spatial structural priors of the SD. Consequently, the DDPM efficiently models the TD distributions within a stable and discriminative latent space.

As illustrated in Phase II of Fig. 1, the initial input feature of the U-Net is denoted as z^t. During the downsampling stage, the input is processed by 3-D convolution layers. To prevent information loss caused by halving the spectral dimension (B) via stride constraints, the network employs an information compensation mechanism: the number of feature channels progressively doubles. Consequently, the feature tensor dimensions sequentially evolve into $32 \times B/2$ and $64 \times B/4$, eventually reaching a $128 \times B/8$ bottleneck layer. This progressive channel expansion effectively suppresses spectral redundancy while projecting the physical spectral information into a wider, high-dimensional semantic latent space. Subsequently, in the upsampling stage, 3-D transposed convolutions are utilized to gradually recover the spectral resolution, and skip connections are introduced to concatenate deep and shallow features of the same scale along the channel dimension. During this process, the tensor dimensions sequentially transform into $64 \times B/4$ and $32 \times B/2$, ultimately restoring to a $16 \times B$ representation before the final output layer.

The detailed training procedure of this latent diffusion is formulated as follows.

1) Forward Spectral–Spatial Diffusion Process: The forward diffusion process progressively corrupts the initial spectral–spatial patch $\boldsymbol{Z}_t = \{z_t\}_{j=1}^{N_t}$ by injecting Gaussian noise over t steps. This procedure follows a Markov chain, where the transition at each diffusion step k depends exclusively on the preceding state z_t^{k-1}:

$$q\left(z_t^k \,|z_t^{k-1}\right) = \mathcal{N}\left(z_t^k;\ \sqrt{1-\beta_k}z_t^{k-1}, \beta_k \boldsymbol{I}\right) \tag{11}$$

where $\beta_\tau \in (0,1)$ dictates the noise variance schedule and $\boldsymbol{I}$ denotes the identity matrix. Defining $\alpha_k = 1 - \beta_k$ and $\bar{\alpha}_k = \prod_{i=1}^{k} \alpha_i$, the noisy latent representation at an arbitrary step k can be marginalized directly from z_0

$$z_t^k = \sqrt{\bar{\alpha}_k} z_t^0 + \sqrt{1-\bar{\alpha}_k}\boldsymbol{\eta} \tag{12}$$

where $\boldsymbol{\eta} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ denotes the true Gaussian noise. This closed-form property avoids the iterative computation of intermediate states.

2) Reverse Spectral–Spatial Diffusion Process: Conversely, the reverse generative process aims to reconstruct the original data z_t^0 from a pure noise state by approximating the intractable reverse transition probabilities. This process is modeled via a neural network p_ϕ parameterized by ϕ

$$p_\phi\left(z_t^{k-1} \,|z_t^k\right) = \mathcal{N}\left(z_t^{k-1}; \boldsymbol{\mu}_\phi\left(z_t^k, k\right), \sigma_k^2 \boldsymbol{I}\right) \tag{13}$$

where the variance σ_k^2 is analytically derived from the forward process posterior

$$\sigma_k^2 = \frac{1-\bar{\alpha}_{k-1}}{1-\bar{\alpha}_k}\beta_k. \tag{14}$$

The parameterized mean $\boldsymbol{\mu}_\phi(z_t^k, k)$ is reformulated to predict the injected noise via a denoiser neural network $\boldsymbol{\eta}_\phi$

$$\boldsymbol{\mu}_\phi\left(z_t^k, k\right) = \frac{1}{\sqrt{\alpha_k}}\left(z_t^k - \frac{1-\alpha_k}{\sqrt{1-\bar{\alpha}_k}}\boldsymbol{\eta}_\phi\left(z_t^k, k\right)\right). \tag{15}$$

The denoiser $\boldsymbol{\eta}_\phi$ employs a 3-D encoder–decoder architecture conditioned on time-step embeddings, enabling the effective modeling of complex spectral–spatial correlations across varying noise levels.

3) Diffusion Loss: For optimization, the model minimizes the L_1 norm between the true Gaussian noise $\boldsymbol{\eta}$ and the predicted noise $\boldsymbol{\eta}_\phi(z_t^k, k)$, providing enhanced robustness compared to the standard mean squared error. The training objective is formulated as the expectation over all time steps and data samples

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\boldsymbol{\eta},k}\left[\left\|\boldsymbol{\eta} - \boldsymbol{\eta}_\phi\left(z_t^k, k\right)\right\|_1\right]. \tag{16}$$

By minimizing the diffusion objective $\mathcal{L}_{\text{diff}}$, the process yields the optimized DDPM parameterized by $\boldsymbol{\eta}_\phi$.

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C. Cycle Training in Phase III

Given that directly classifying transitional representations generated by the DDPM may lead to unstable adaptation across intermediate domains, we introduce the SSBCE cyclic training paradigm, which alternates the SA subphase and the BA subphase to progressively refine cross-domain alignment.

1) Overview of Manifold Evolution: Since the DDPM is trained exclusively on TD latent features in Phase II, it learns a target-aware denoising prior. Under the standard noise-prediction parameterization, the predicted noise can be interpreted as a surrogate of the TD score field

$$s_\phi(z,k) \approx -\frac{\eta_\phi(z^k,k)}{\sqrt{1-\bar{\alpha}_k}} \approx \nabla_z \log P_k^{(t)}(z) \tag{17}$$

where z denotes a generic latent feature variable.

For the SD adaptation, source representations are first partially perturbed to an intermediate timestep k_0. Let $\tilde{P}_k^{(s)}$ and $P_k^{(t)}$ denote the forward-diffused SD and TD distributions, respectively. Under the continuous-time approximation, their KL divergence evolves as follows:

$$\frac{\partial}{\partial k} D_{\mathrm{KL}}\left(\tilde{P}_k^{(s)} \middle\| P_k^{(t)}\right) = -\frac{1}{2} g(k)^2 \, \mathbb{E}_{\tilde{P}_k^{(s)}} \left[\left\| \nabla_z \log \tilde{P}_k^{(s)}(z) - \nabla_z \log P_k^{(t)}(z) \right\|_2^2 \right] \tag{18}$$

where $g(k)$ denotes the scalar diffusion coefficient of the forward SDEs. This formulation indicates that the KL divergence decreases along the forward diffusion process, as Gaussian perturbation progressively smooths domain-specific discrepancies. Assuming that the smoothed target distribution $P_k^{(t)}$ satisfies the logarithmic Sobolev inequality (LSI) with a positive constant $C_{\mathrm{LSI}}(k)$, the corresponding contraction from 0 to k_0 can be bounded by

$$D_{\mathrm{KL}}\left(\tilde{P}_{k_0}^{(s)} \middle\| P_{k_0}^{(t)}\right) \le \exp\left(-\int_0^{k_0} C_{\mathrm{LSI}}(k)\, g(k)^2 \, dk\right) \times D_{\mathrm{KL}}\left(\tilde{P}_0^{(s)} \middle\| P_0^{(t)}\right). \tag{19}$$

Therefore, forward perturbation suppresses part of the source-specific characteristics and reduces the distributional discrepancy at the noisy state. The subsequent reverse process, parameterized by the TD-trained DDPM, denoises the perturbed SD representations under the learned target prior, yielding transitional features that retain source semantics while incorporating TD characteristics.

For continuous-time interpretation, the corresponding probability flow ordinary differential equation is

$$dz = \left[f(z,k) - \frac{1}{2} g(k)^2 \nabla_z \log P_k^{(t)}(z) \right] dk. \tag{20}$$

This formulation provides a theoretical view of target-guided reverse transport, while the actual Diff-MEA implementation employs the discrete DDPM reverse process. By progressively increasing the diffusion timestep k during cyclic training, a sequence of transitional representations with increasing TD characteristics is generated to support the subsequent cyclic optimization in the SSBCE.

2) Synergistic Semantic-Boundary Coevolution: Building upon the diffusion-induced transitional representations, the SSBCE alternates SA and BA across a sequence of intermediate manifolds parameterized by progressively increasing diffusion timesteps k.

Within the cyclic training paradigm, the manifold transition evolves progressively by enforcing a strict synchronization between the cyclic iteration index n and the diffusion timestep k. At a given training epoch associated with timestep k, the source latent features $\boldsymbol{Z}_s = \{z_s\}_{j=1}^{N_s}$ are first perturbed through k forward diffusion steps according to (12), producing the noisy intermediate representation $\boldsymbol{Z}_s^k = \{z_s^k\}_{j=1}^{N_s}$. Subsequently, the reverse diffusion process defined in (13) employs the target-aware denoiser to generate a timestep-specific transitional feature $\tilde{\boldsymbol{Z}}^k = \{\tilde{z}_{tran}^k\}_{j=1}^{N_s}$. The generation process is formulated as follows:

$$\tilde{z}_{tran}^k = \frac{1}{\sqrt{\bar{\alpha}_k}} \left(z_s^k - \sqrt{1-\bar{\alpha}_k}\, \eta_\phi^k\left(z_t^k, k\right) \right). \tag{21}$$

This partial noise–denoise transformation progressively transports source-biased features toward the target distribution, yielding a microshifted intermediate domain state while preserving structural semantics.

To optimize the model along the evolutionary trajectory, the SSBCE alternates between two complementary subphases at each transitional state $\tilde{\boldsymbol{Z}}^k$. In the SA subphase, the MTDA is fixed while the DDPM is optimized to preserve source-domain semantics during distribution evolution. In the subsequent BA subphase, the DDPM is frozen, and the MTDA is updated on $\tilde{\boldsymbol{Z}}^k$ to progressively adapt the decision boundary to the evolving manifold.

After completing the dual optimization at timestep k, the diffusion timestep is monotonically increased ($k = k + 1$) to initiate the next training cycle. Through this alternating procedure, the framework progressively constructs a discrete sequence of transitional features $\{\tilde{\boldsymbol{Z}}^i\}_{i=1}^t = \{\tilde{\boldsymbol{Z}}^1, \tilde{\boldsymbol{Z}}^2, \ldots, \tilde{\boldsymbol{Z}}^t\}$ through (21), where t signifies the total number of diffusion timesteps. By repeatedly generating adjacent transitional states such as $\tilde{\boldsymbol{Z}}^{k+1}$, the cyclic optimization compels both the generator and the classifier to adapt to fine-grained distributional shifts between consecutive timesteps (k and $k + 1$). As the cyclic training progresses iteratively from $k = 1$ to $k = t$, the network parameters undergo a synchronized coevolution. The DDPM parameters evolve along the trajectory $\{\eta_\phi^1, \eta_\phi^2, \ldots, \eta_\phi^k, \ldots\}$ during SA, enforcing topology-preserving generation and maintaining class-discriminative structures. Concurrently, the MTDA parameters are iteratively updated as $\{\boldsymbol{M}^1, \boldsymbol{M}^2, \ldots, \boldsymbol{M}^k, \ldots\}$, which enables progressive assimilation of the nonconvex target manifold. During cyclic optimization, the MTDA state exhibiting the most favorable training objective is retained as $\boldsymbol{M}^\star$. Through this coupled optimization process, the classifier dynamically expands its decision boundaries while mitigating catastrophic forgetting, ultimately achieving stable and smooth cross-domain adaptation.

3) Subphase (SA): The objective of the SA subphase is to regulate the probabilistic transport induced by the DDPM, thereby preventing catastrophic semantic drift that may arise during continuous manifold deformation. As the diffusion

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model aligns source data toward the target distribution, uncontrolled adaptation may distort interclass decision boundaries. To mitigate this risk, we employ an asymmetric optimization strategy in which the MTDA module remains strictly frozen while only the DDPM parameters are updated.

Under this configuration, the frozen MTDA serves as a domain-invariant semantic evaluator. During the forward pass, the intermediate representations generated by the DDPM are fed directly into the MTDA. During backpropagation, gradients propagate through the fixed MTDA network without modifying its parameters and are applied exclusively to the DDPM. This design constrains the generative dynamics to remain consistent with established categorical boundaries, forcing the diffusion trajectory to preserve semantic structures while traversing the nonconvex feature manifold.

Mathematically, the DDPM is optimized via a joint SA loss

$$\mathcal{L}_{sa} = \mathcal{L}_{\text{diff}} + \lambda_a \mathcal{L}_{\text{SC}}^{(k)} \tag{22}$$

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\boldsymbol{\eta},k}\left[\left\|\boldsymbol{\eta} - \boldsymbol{\eta}_{\phi}^{k}\left(\boldsymbol{z}_t^k, k\right)\right\|_1\right] \tag{23}$$

$$\mathcal{L}_{\text{SC}}^{(k)} = -\frac{1}{M}\sum_{i=1}^{M}\sum_{c=1}^{N_c} y_{i,c}^{s} \log\left(\ M^k\left(\tilde{\boldsymbol{Z}}_i\right)\right)_c \tag{24}$$

where $\mathcal{L}_{SC}^{(k)}$ represents the standard cross-entropy loss evaluated at the kth transitional state, λ_a denotes the balancing coefficient, and M denotes the number of samples in a mini-batch. The diffusion loss $\mathcal{L}_{\text{diff}}$ drives the generative process by encouraging the DDPM to produce features aligned with the target-domain distribution rather than reverting toward the source manifold. Conversely, the semantic consistency term $\mathcal{L}_{\text{SC}}^{(k)}$ imposes an explicit topological constraint that preserves class-level semantics. By enforcing agreement with the ground-truth source labels y^s, this term penalizes deviations from the original categorical structure during generative transport. Through this joint optimization, the DDPM learns to generate semantically stabilized transitional representations that remain class-discriminative while progressively approaching the target distribution. As a result, the SA subphase establishes a stable and well-separated intermediate manifold, providing a reliable foundation for the next BA subphase.

4) Subphase (BA): Following the SA of the feature space, the intermediate representations have progressively shifted toward the target distribution. The primary objective of this subphase is to dynamically recalibrate the decision boundaries of the MTDA to accurately fit this newly evolved, nonconvex manifold. To achieve this without disrupting the established generative trajectory, we invert the asymmetric optimization paradigm: the DDPM parameters are strictly frozen, while the MTDA is actively updated. During backpropagation, gradients derived from the task-specific loss traverse the MTDA to update the memory tokens and network weights, explicitly terminating before the diffusion model. This targeted gradient routing enables the classifier to adapt to the shifted topological structures without perturbing the latent data generation process.

In particular, rather than producing a single final set, the frozen DDPM iteratively generates transitional features at each sequential noise step, yielding a trajectory sequence

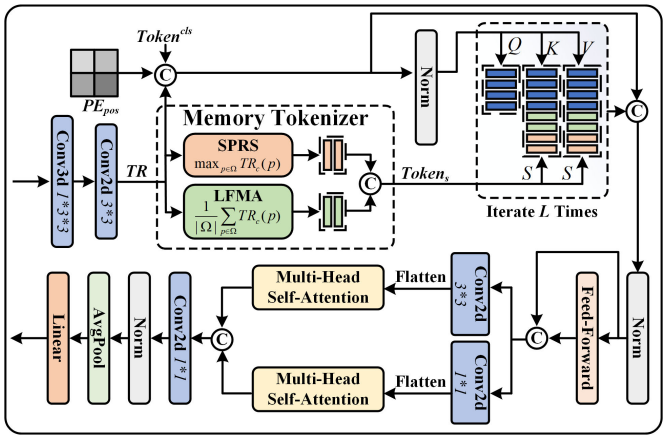


Fig. 2. Detailed framework of the proposed MTDA.

$\{\tilde{\boldsymbol{Z}}^i\}_{i=1}^k = \{\tilde{\boldsymbol{Z}}^1, \tilde{\boldsymbol{Z}}^2, \ldots, \tilde{\boldsymbol{Z}}^k\}$. To facilitate robust boundary transfer, we construct a composite training set $\mathcal{Z}_{\text{mix}} = \bigcup_{n=1}^{k}\{\tilde{\boldsymbol{Z}}^n\}$ that encompasses the representations across all intermediate transitional levels. The MTDA is subsequently optimized by minimizing the following objective:

$$\mathcal{L}_{ba} = \mathcal{L}_{\text{SC}}^{(k)} + \sum_{n=1}^{k-1} \lambda_k \mathcal{L}_{\text{SC}}^{(n)} \tag{25}$$

where $\lambda_k = 1/k$ denotes the step-dependent decay coefficient. This monotonically decreasing weighting scheme establishes a curriculum learning strategy that prioritizes the transitional features $\tilde{\boldsymbol{Z}}^k$ exhibiting a higher degree of target alignment at the current step t while concurrently maintaining a residual rehearsal of the source-dominant information from preceding stages. Consequently, this mechanism effectively mitigates catastrophic forgetting throughout the continuous domain transition. By dynamically calibrating its decision boundaries to the evolving manifold, the MTDA sustains optimal semantic discriminability, thereby serving as a robust semantic anchor for the subsequent DDPM optimization cycle.

D. Final Test in Phase IV

Upon completion of the SSBCE cyclic optimization, the optimized MTDA parameters $\boldsymbol{M}^{\star}$ retained during training are integrated with the frozen Mamba-based encoder f_{ψ} to construct the final domain-adaptive inference model. The resulting network directly performs prediction on the unlabeled TD samples without any additional parameter updating or fine-tuning. By exploiting the progressively evolved transitional manifolds and the recalibrated decision boundaries established during training, the resulting model effectively accommodates substantial cross-domain discrepancies while preserving class-discriminative semantics.

IV. Experimental Results and Analysis

A. Data Description

1) Houston Dataset: The Houston dataset comprises the source scene Houston13 (H13) and the target scene Houston18 (H18), acquired over the University of Houston in June

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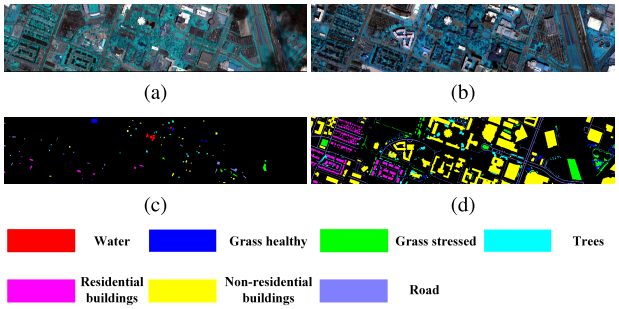


Fig. 3. Houston dataset visualization. (a) Houston2013 false-color image. (b) Houston2018 false-color image. (c) Houston2013 ground-truth. (d) Houston2018 ground-truth.

TABLE I
SAMPLE DISTRIBUTION OF HOUSTON DATASET SCENES: HOUSTON13 AND HOUSTON18

Class	Name	Houston13	Houston18
C1	Grass healthy	345	1353
C2	Grass stressed	365	4888
C3	Trees	365	2766
C4	Water	285	22
C5	Residential buildings	319	5347
C6	Non-residential buildings	408	32459
C7	Road	443	6365
Total		**2530**	**53200**

TABLE II
SAMPLE DISTRIBUTION OF SH2HZ DATASET SCENES: SHANGHAI AND HANGZHOU

Class	Name	Shanghai	HangZhou
C1	Water	123123	18043
C2	Land/Building	161689	77450
C3	Plant	83188	40307
Total		**368000**	**135800**

2012 (CASI-1500 sensor) and February 2017 (1-m resolution, 48 bands), respectively. Despite sharing identical wavelength ranges, the temporal and spatial discrepancies between the scenes induce a severe domain shift. To achieve domain alignment, a 209×955 subregion of H13 was extracted to spatially match H18, and its spectral channels were adjusted to 48 bands. As detailed in Table I, this task involves seven shared land-cover categories, containing 2530 labeled source samples and 53 200 unlabeled target samples. This setup establishes a realistic benchmark for hyperspectral UDA. False-color images and ground-truth maps are illustrated in Fig. 3.

2) SH2HZ Dataset: The Shanghai–Hangzhou (SH2HZ) dataset consists of two EO-1 Hyperion scenes, representing the SD Shanghai and the TD Hangzhou. Following the removal of noisy and uncalibrated spectral channels, each scene retains 198 valid bands. The spatial dimensions of the Shanghai and Hangzhou tiles are 1600×230 and 590×230 pixels, respectively. Both scenes share three common land-cover categories utilized for cross-scene classification, as summarized in Table II. False-color composites and corresponding ground-truth maps are illustrated in Fig. 4.

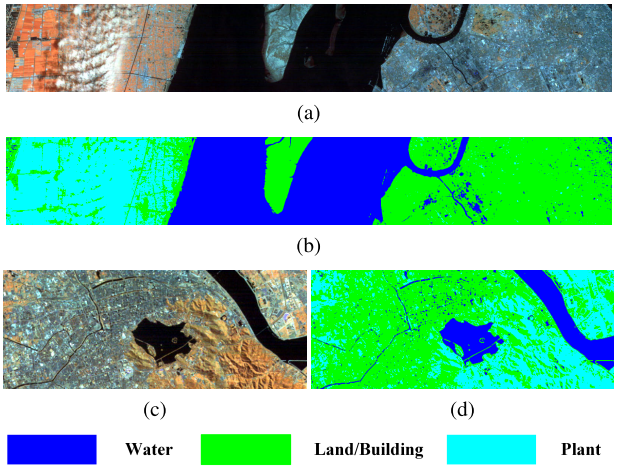


Fig. 4. SH2HZ dataset visualization. (a) Shanghai false-color image. (b) Shanghai ground-truth. (c) Hangzhou false-color image. (d) Hangzhou ground-truth.

TABLE III
SAMPLE DISTRIBUTION OF INDIANA DATASET SCENES: SOURCE AND TARGET

Class	Name	Source scene	Target scene
C1	Concrete/Asphalt	4867	2942
C2	Corn cleanTill	9822	6029
C3	Corn cleanTill EW	11414	7999
C4	Orchard	5106	1562
C5	Soybeans cleanTill	4731	4792
C6	Soybeans cleanTill EW	2996	1638
C7	Wheat	3223	10739
Total		**42159**	**35701**

3) Indiana Dataset: The Indiana dataset was acquired by the AVIRIS sensor over northwestern Tippecanoe County, Indiana, in 1992, at a nominal spatial resolution of 20 m. The original scene is partitioned into two nonoverlapping subsets serving as the source and target domains. Each subset measures 400×300 pixels and comprises 220 spectral bands. As detailed in Table III, both domains encompass seven shared vegetation-related classes, providing an evaluation scenario for cross-domain classification under spectral–spatial variability. False-color composites and corresponding ground-truth maps are illustrated in Fig. 5.

B. Experimental Settings

To evaluate the proposed approach, we compare it against seven state-of-the-art domain adaptation methods: DSAN [26], TSTnet [25], SCLUDA [27], MLUDA [28], MSDA [29], S^4DL [31], and S2PST [30]. To simulate practical limited-annotation scenarios, the training set comprises 30 randomly sampled labeled instances per class from the SD alongside all available unlabeled data from the TD. Baseline hyperparameters are initialized to their original configurations and subsequently fine-tuned to ensure fair comparison. To mitigate variance induced by random initialization, all experiments are independently repeated ten times, reporting the average results.

The network parameters are iteratively optimized using the Adam algorithm across a three-phase training pipeline. In

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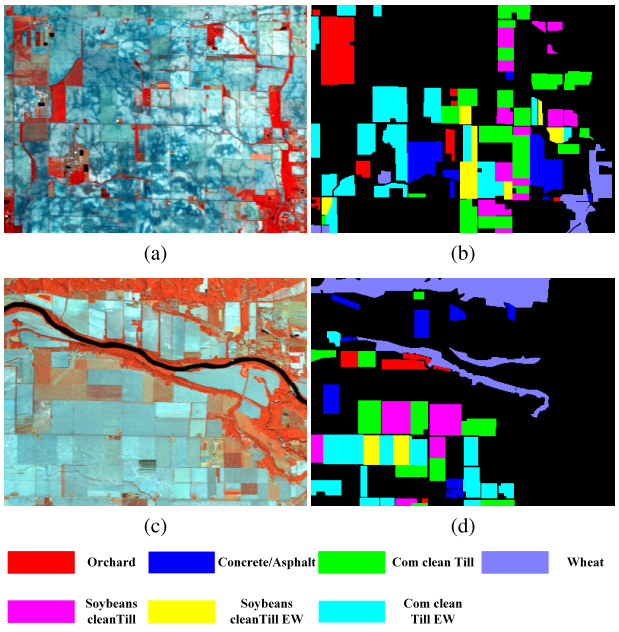


Fig. 5. Indiana dataset visualization. (a) Source false-color image. (b) Source ground-truth map. (c) Target false-color image. (d) Target ground-truth map.

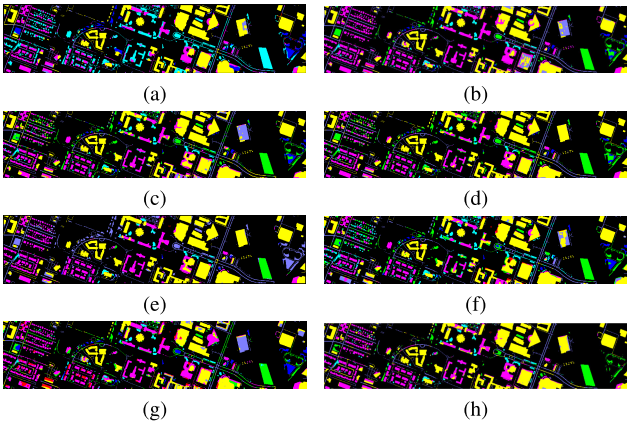


Fig. 6. Classification maps obtained via various algorithms for the Houston dataset. (a) DSAN. (b) TSTnet. (c) SCLUDA. (d) MLUDA. (e) MSDA. (f) S2PST. (g) S^{4}DL. (h) Diff-MEA.

Phase I, the model is pretrained on the SD for 200 epochs with a batch size of 32; the learning rate is set to 1×10^{-4} for the Houston and SH2HZ datasets, and 5×10^{-4} for the Indiana dataset. Phase II trains the DDPM on the TD for 100 epochs, utilizing $t = 600$ diffusion timesteps, a batch size of 32, and a learning rate of 1×10^{-4}. During Phase III (cyclic training), the batch size is increased to 64, and the learning rate is adjusted to 1×10^{-4} for Houston and SH2HZ, and 1×10^{-3} for Indiana. The loss balancing parameter λ_a is set to 1 for Houston and SH2HZ, and 10 for Indiana.

Classification performance is quantitatively evaluated using overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ). Systematic ablation studies are then conducted to isolate the contributions of individual network components, followed by a comprehensive sensitivity analysis assessing model robustness to four critical hyperparameters: the input patch size, the balancing coefficient λ_a, the number of 3-D convolutional layers in the DDPM, and the latent space feature dimension C. Finally, the computational efficiency of the proposed method is assessed by comparing its trainable parameters, computational complexity (FLOPs), and inference time against the baseline models.

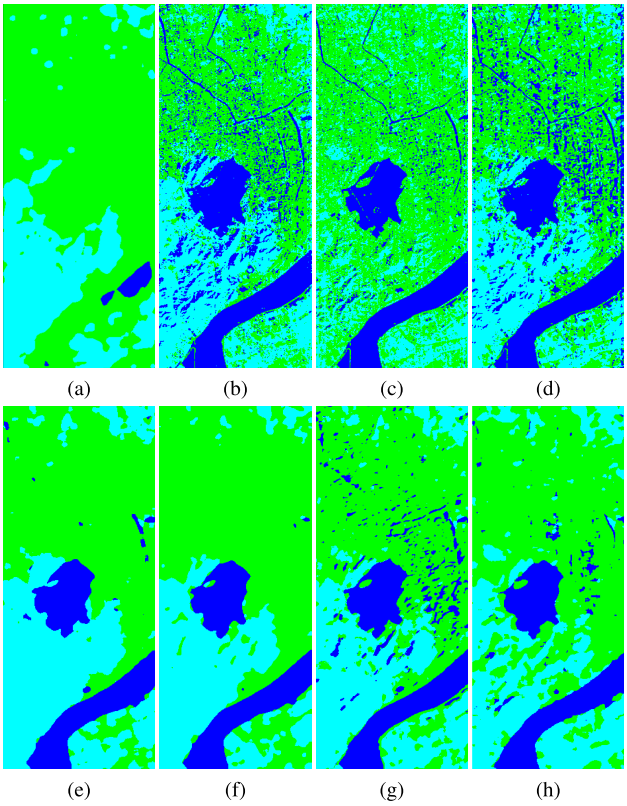


Fig. 7. Classification maps obtained via various algorithms for the SH2HZ dataset. (a) DSAN. (b) TSTnet. (c) SCLUDA. (d) MLUDA. (e) MSDA. (f) S2PST. (g) S^{4}DL. (h) Diff-MEA.

C. Comparing With Other Methods

The experimental results are summarized in Tables IV–VI, with the top-performing outcomes highlighted in bold. To facilitate visual comparisons, the best classification maps produced by different methods are exhibited in Figs. 6–8.

1) Experiments on Houston Task: Table IV presents classification results on the Houston dataset. Notably, the proposed Diff-MEA achieves the highest OA of 73.60%, AA of 75.42%, and κ of 59.60%, surpassing the second-best approaches by 1.41%, 2.75%, and 3.03%, respectively. Furthermore, the Diff-MEA exhibits the lowest performance variance (e.g., $\pm$0.64% in OA), markedly lower than the fluctuations observed in the baseline methods. Analytically, the proposed framework effectively mitigates the inherent tradeoff between global distribution alignment and local semantic preservation under large domain shifts. While contemporary methods such as MLUDA and S2PST achieve competitive aggregate metrics, their reliance on single-step hard alignment frequently induces feature representation distortion, thereby degrading the classification accuracy of easily confused classes. By reformulating UDA as a continuous manifold evolution process, the proposed SSBCE mechanism provides a smooth transition across the domain gap. Consequently, Diff-MEA demonstrates robust

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TABLE IV
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE HOUSTON DATASET

Methods	DSAN [26]	TSTnet [25]	SCLUDA [27]	MLUDA [28]	MSDA [29]	S2PST [30]	S^{4}DL [31]	Diff-MEA
C1	73.36±18.04	53.34±5.02	82.99±6.88	53.66±5.08	**94.08±3.62**	81.41±8.45	63.43±4.51	65.29±15.23
C2	36.78±13.79	74.46±15.95	69.55±7.74	73.86±10.28	53.09±8.21	72.94±7.00	77.63±23.05	**79.08±3.87**
C3	65.21±3.80	47.26±21.88	67.79±4.10	44.75±9.61	56.39±18.95	66.26±5.30	57.46±11.54	**72.80±6.21**
C4	60.60±3.24	9.09±21.89	80.91±17.34	71.82±22.71	50.45±16.45	**93.64±6.14**	91.82±7.96	85.13±23.38
C5	54.54±26.74	**94.28±2.18**	84.97±5.46	77.22±8.40	81.96±8.12	64.92±7.03	70.48±11.56	88.65±7.84
C6	75.80±2.82	67.03±8.86	61.82±5.99	71.17±7.50	78.91±7.54	65.88±5.19	**80.83±5.59**	72.50±3.00
C7	27.56±21.73	36.49±5.24	58.45±14.08	57.59±25.33	46.63±14.14	63.65±6.02	29.91±17.73	**64.46±8.47**
OA	63.69±1.94	65.40±4.89	65.31±3.05	68.58±3.66	72.19±5.66	66.59±2.73	71.75±2.19	**73.60±0.64**
AA	56.26±4.01	54.56±5.12	72.35±3.42	64.30±7.57	65.93±5.76	72.67±1.73	67.37±6.29	**75.42±2.58**
$\kappa \times 100$	44.74±3.28	48.24±5.67	50.64±3.24	52.66±4.72	56.45±7.08	52.32±2.51	56.57±5.15	**59.60±0.95**

TABLE V
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE SH2HZ DATASET

Methods	DSAN [26]	TSTnet [25]	SCLUDA [27]	MLUDA [28]	MSDA [29]	S2PST [30]	S^{4}DL [31]	Diff-MEA
C1	84.56±3.55	83.19±13.12	**98.94±1.24**	93.07±1.42	88.00±1.83	86.30±1.02	71.06±1.63	88.50±6.23
C2	86.41±2.45	81.67±8.23	71.19±8.64	77.72±5.97	77.53±5.88	**87.50±2.62**	84.73±2.81	83.16±0.99
C3	56.15±8.14	66.15±8.15	78.79±13.01	75.19±7.41	**79.39±6.03**	62.74±6.08	76.15±4.56	79.21±1.95
OA	77.18±1.19	77.27±2.18	77.13±6.08	79.01±3.80	79.47±1.75	79.99±1.12	80.37±1.13	**82.70±1.00**
AA	75.71±1.36	77.00±5.00	82.97±4.27	81.99±2.87	81.64±0.70	78.85±1.34	77.31±1.56	**83.62±0.28**
$\kappa \times 100$	60.56±0.23	60.72±6.25	60.47±8.47	63.72±5.68	64.52±2.18	65.42±1.90	66.07±1.49	**70.10±1.09**

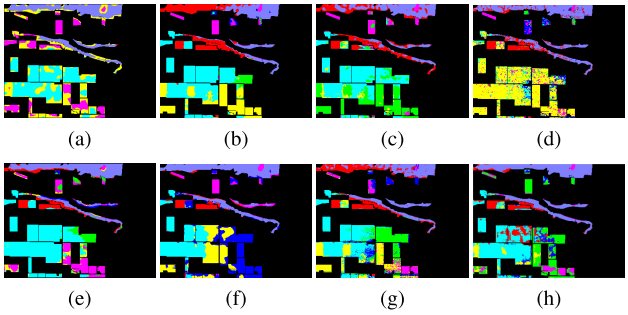


Fig. 8. Classification maps obtained via various algorithms for the Indiana dataset. (a) DSAN. (b) TSTnet. (c) SCLUDA. (d) MLUDA. (e) MSDA. (f) S2PST. (g) S^{4}DL. (h) Diff-MEA.

discrimination on complex categories characterized by severe spectral confusion and class imbalance, delineating clearer decision boundaries for challenging classes such as "Grass stressed" (C2), "Trees" (C3), and "Road" (C7), where it attains peak accuracies of 79.08%, 72.80%, and 64.46%, respectively.

2) Experiments on SH2HZ Task: Table V presents the cross-scene classification performance on the SH2HZ dataset. The proposed Diff-MEA achieves state-of-the-art results, recording the OA of 82.70%, AA of 83.62%, and κ of 70.10%.

The SH2HZ dataset introduces specific challenges regarding the spatial distribution of the "Water" class (C1), which comprises both extensive continuous bodies and fragmented river networks. Although methods such as SCLUDA, MLUDA, and MSDA report high accuracy on C1, qualitative analysis (Fig. 7) indicates that this sensitivity is accompanied by a high false-positive rate, systematically misclassifying substantial regions of "Land/Building" and "Plant" as "Water" and producing irregular spatial artifacts. On the contrary, the Diff-MEA enforces macroscopic topological coherence. By utilizing LFMA and SPRS to synergistically extract regional contextual information and acute discriminative cues, the model accurately delineates primary water bodies while suppressing scattered false positives. While this approach necessitates a minor tradeoff in sensitivity for highly narrow river segments, it substantially preserves the overall structural integrity and visual coherence.

Although the Diff-MEA achieves higher overall metrics, deterministic baselines such as MLUDA and SCLUDA occasionally outperform it on simple categories with distinct spectral signatures (e.g., "Water" in the SH2HZ dataset). This discrepancy arises from fundamentally different alignment mechanisms. Deterministic hard alignment effectively unifies separable classes but frequently distorts the nonconvex boundaries of entangled hard classes, which causes topological collapse. On the contrary, the probabilistic manifold evolution of Diff-MEA injects controlled stochasticity. This stochasticity slightly relaxes the rigid boundaries of simple classes, resulting in marginal accuracy degradation. Critically, it also preserves the underlying topological structure, thereby protecting the fragile interclass margins among confused classes (e.g., "Grass stressed," "Road" in Houston dataset) and significantly boosting their accuracy. Consequently, Diff-MEA prioritizes globally faithful topological mapping over locally overfitting simple classes.

3) Experiments on Indiana Task: The Indiana dataset serves as a highly challenging cross-scene benchmark due to severe domain shifts and complex spectral mixing among structurally similar species. Despite such extreme spectral complexity, the Diff-MEA establishes a clear quantitative dominance. As reported in Table VI, the Diff-MEA achieves superior

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TABLE VI
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE INDIANA DATASET

Methods	DSAN [26]	TSTnet [25]	SCLUDA [27]	MLUDA [28]	MSDA [29]	S2PST [30]	S^{4}DL [31]	Diff-MEA
C1	23.64±15.49	**26.81±6.36**	5.36±5.63	6.58±3.61	0.67±1.19	11.38±6.66	6.18±7.69	11.48±10.76
C2	0.83±0.77	1.06±2.24	9.25±18.43	**26.94±12.97**	21.38±12.87	8.29±9.40	22.74±7.89	13.09±13.33
C3	53.08±11.23	19.27±4.62	23.25±15.75	28.80±7.05	**57.19±19.16**	40.40±6.68	43.64±9.26	50.85±7.52
C4	61.66±11.86	16.69±5.15	64.18±11.09	87.02±3.98	**95.96±6.45**	81.47±13.61	45.59±3.94	88.39±7.06
C5	22.68±8.27	20.32±4.63	**31.75±22.90**	16.51±7.19	7.02±7.39	0.96±1.19	6.96±5.61	2.49±7.89
C6	18.76±11.99	13.79±4.39	30.59±20.28	18.87±13.10	13.08±30.35	2.60±8.05	**32.17±18.78**	8.80±9.39
C7	60.54±12.12	**95.08±3.61**	82.41±5.72	72.35±10.03	61.79±24.27	84.70±3.99	73.05±7.11	84.98±10.39
OA	38.79±1.02	39.40±3.61	40.48±3.10	40.20±3.61	40.81±2.59	40.68±0.75	40.51±1.08	**44.72±1.12**
AA	34.46±3.01	27.57±2.92	35.26±1.98	36.72±1.37	36.73±4.07	32.83±2.45	32.90±1.68	**37.15±1.59**
$\kappa \times 100$	24.32±1.54	24.98±3.39	27.97±8.23	28.07±3.33	29.87±3.73	30.64±1.13	31.45±1.04	**33.07±2.54**

performance across all core metrics, specifically reaching OA of 44.72%, AA of 37.15%, and κ of 33.07. Compared with the second-best method MSDA (OA of 40.81%), the Diff-MEA achieves a significant lead of 3.91% in terms of OA. Crucially, while MSDA generates outputs with higher localized visual smoothness, this spatial homogenization obscures systematic biases in large-scale land-cover distributions, frequently resulting in the erroneous classification of entire agricultural plots into adjacent categories. Conversely, although Diff-MEA yields occasional spatially fragmented misclassifications, it accurately maps the macroscopic geographical extent of disparate crop types. These findings indicate that when optimizing across highly confused and nonconvex target manifolds, Diff-MEA preserves the authenticity of semantic boundaries and avoids the oversmoothing degradation inherent to conventional spatial regularization techniques.

TABLE VII
ABLATION STUDY OF THE PROPOSED DIFF-MEA ON THREE DATASETS

Dataset	**Case**	**OA**(%)	**AA**(%)	$\kappa \times 100$
Houston	*Case 1*	65.67 ± 2.16	61.02 ± 5.19	50.91 ± 2.52
	Case 2	67.97 ± 1.22	63.62 ± 7.02	53.23 ± 1.49
	Case 3	68.34 ± 0.36	69.97 ± 0.98	55.91 ± 0.41
	Case 4	**73.60 ± 0.64**	**75.42 ± 2.58**	**59.60 ± 0.95**
SH2HZ	*Case 1*	71.59 ± 0.92	79.05 ± 2.53	54.82 ± 2.26
	Case 2	76.45 ± 1.69	79.12 ± 4.23	60.19 ± 4.22
	Case 3	77.39 ± 0.10	79.47 ± 0.02	61.66 ± 0.13
	Case 4	**82.70 ± 1.00**	**83.62 ± 0.28**	**70.10 ± 1.09**
Indiana	*Case 1*	35.03 ± 1.56	30.81 ± 0.93	21.79 ± 1.48
	Case 2	36.31 ± 1.88	31.26 ± 1.86	23.45 ± 1.78
	Case 3	37.85 ± 0.28	32.28 ± 0.16	24.47 ± 0.23
	Case 4	**44.72 ± 1.12**	**37.15 ± 1.59**	**33.07 ± 2.54**

D. Ablations on the Diff-MEA

To validate the effectiveness of each component within the proposed Diff-MEA, we conduct a comprehensive ablation study focusing on its core modules and optimization schemes. This analysis is designed to systematically isolate and quantify three primary methodological contributions to the overall classification performance. The four experimental configurations are formulated as follows.

Case 1: The MTDA module is excluded and replaced with a standard linear classifier. This baseline configuration employs an end-to-end training strategy to execute single-step hard domain alignment, utilizing the SA subphase solely while completely omitting the alternating training paradigm.

Case 2: Building upon Case 1, this variant implements a multistage training strategy that encompasses Phases I through IV. However, the cyclic alternating training mechanism within Phase III remains excluded.

Case 3: Building upon Case 2, this configuration incorporates the SSBCE cyclic training mechanism, which strictly alternates between the SA subphase and the BA subphase.

Case 4: The proposed MTDA module is introduced to replace the linear classifier, constituting the complete Diff-MEA framework configuration.

As shown in Table VII, classification performance consistently improves as additional components are incorporated. Initially, introducing the multistage training strategy increases OA by 4.86% and 2.30% on SH2HZ and Houston, respectively, demonstrating that decoupling feature extraction from adaptation alleviates semantic drift caused by aggressive single-stage alignment. By strictly freezing the encoder, we prevent $\mathcal{L}_{diff}$ from distorting established semantic structures and mitigate catastrophic forgetting. This freezing strategy establishes a fixed coordinate basis that preserves transferable primitives, ensuring the target probability law remains intact. As a result, the DDPM can focus exclusively on modeling the noised target distribution $P_k^{(t)}$. By learning a target-aware denoising prior over $P_k^{(t)}(z)$, the parameterized denoiser induces a directional field that progressively transports transitional features toward high-density target regions. Subsequently, the SSBCE cyclic mechanism further enhances training stability, improving Houston's AA by 6.35% while reducing the OA standard deviation to 0.36%. By alternating the SA and the BA subphases, the SSBCE decomposes large domain discrepancies into manageable adaptation steps, effectively preserving feature topology and preventing representation collapse. Finally, the MTDA delivers the most pronounced gains, boosting OA by 5.26% and 6.87% on Houston and Indiana. By synergizing SPRS, LFMA, and decoupled spatial–spectral

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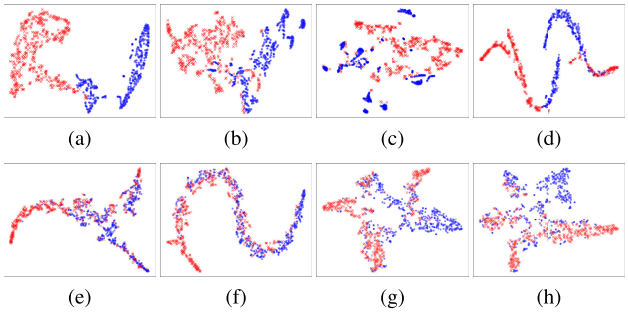


Fig. 9. T-SNE visualizations of feature alignment between the SD and TD for specific classes. (a)–(c) Classes 1, 3, and 6 of the Houston dataset. (d)–(f) Classes 1, 2, and 3 of the SH2HZ dataset. (g) and (h) Classes 5 and 6 of the Indiana dataset.

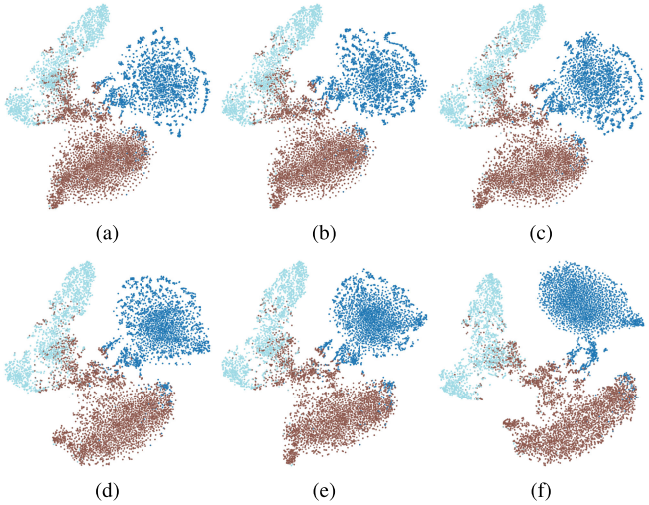


Fig. 10. T-SNE visualizations illustrating how distinct transition domain attributes are iteratively built at selected diffusion timesteps k during the SSBCE cyclic training process on the SH2HZ dataset. (a) $k = 0$. (b) $k = 25$. (c) $k = 50$. (d) $k = 75$. (e) $k = 100$. (f) $k = 200$.

branches, the MTDA's memory tokens act as robust semantic anchors that dynamically integrate spatial context to accurately expand decision boundaries.

E. Analysis of Transition-Domain Feature Evolution

Moreover, we employ t-SNE visualizations in Figs. 9 and 10 to evaluate the effectiveness of the SSBCE from two complementary perspectives. As for the first perspective, Fig. 9 presents the TD features alongside the final transitional features (at the optimal classification stage) for specific classes. Within each selected class, the transition-domain features are closely aligned with the corresponding TD features in the embedding space. This intraclass consistency indicates that the SSBCE preserves class-specific semantic identity during manifold evolution and provides a reliable feature basis for subsequent classifier refinement. Therefore, the visualizations qualitatively demonstrate that the SSBCE successfully preserves semantic identity during the continuous manifold evolution.

From another perspective, Fig. 10 plots the t-SNE visualizations of the transitional domain features generated at different noise steps k during the SSBCE cyclic training process on the SH2HZ dataset. As k increases incrementally, the perturbed source features are continuously pulled toward the high-density TD manifold by the pretrained geometric prior, forming increasingly tight and separable clusters. This progression explicitly demonstrates that the diffusion denoising process under the SSBCE framework ensures the high-fidelity reconstruction of spectral–spatial feature structures, fully addressing concerns regarding the loss of spectral details.

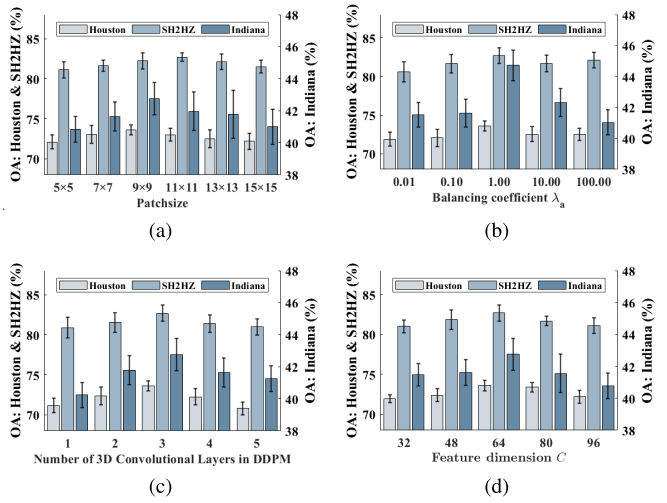


Fig. 11. Sensitivity analysis of parameters. (a) Patch size. (b) λ_a. (c) Number of 3-D convolutional layers in DDPM. (d) Feature dimension C.

F. Parameter and Sensitivity Analysis

To systematically evaluate the robustness of the proposed Diff-MEA framework and the influence of its core hyperparameters on classification performance, a comprehensive parameter sensitivity analysis is conducted. In particular, the investigation focuses on the input patch size, the number of 3-D convolutional layers in the DDPM, the balancing coefficient λ_a in (22), and the feature dimension C of the latent space. During the analysis, only the target parameter was varied, while the remaining parameters were fixed at their optimal settings. The OA serves as the primary metric for evaluating the impact of these variations.

1) Patch size: The influence of the patch size was examined across six configurations: $\{5, 7, 9, 11, 13, 15\}$. As illustrated in Fig. 11(a), the SH2HZ dataset demonstrates a positive correlation between OA and patch size, peaking at 11. Conversely, the Houston and Indiana datasets exhibit an inverted U-shaped trend, achieving optimal performance at a patch size of 9. The influence of patch size was more pronounced for the Houston and SH2HZ datasets compared to the Indiana dataset. Conversely, the Indiana dataset, characterized by strong spectral dominance and high patch-wise consistency, exhibited less sensitivity to patch size variations. However, performance drops with excessively large patches because the DDPM struggles with high-dimensional spatial inputs. Oversized receptive fields tend to incorporate extraneous background or cross-category pixels, diluting semantic purity, while the expanded dimensionality complicates target score function estimation.

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TABLE VIII
COMPARISON OF MODEL COMPLEXITY AND TEST TIME ON THREE DATASETS

Methods	Houston			SH2HZ			Indiana		
	Params(M)	FLOPs(M)	Time(s)	Params(M)	FLOPs(M)	Time(s)	Params(M)	FLOPs(M)	Time(s)
DSAN [26]	0.431	37.52	15.15	0.517	56.96	32.14	0.531	59.81	16.21
TSTnet [25]	5.328	8.51	14.37	5.370	12.82	35.89	5.377	13.46	22.23
SCLUDA [27]	0.254	15.19	16.57	0.602	1.45	37.56	0.654	70.72	21.01
MLUDA [28]	2.104	62.78	15.01	2.451	7.82	31.85	2.505	8.41	19.46
MSDA [29]	0.160	51.39	14.23	0.394	95.40	29.34	0.428	157.55	18.68
S2PST [30]	0.503	185.68	16.11	0.504	187.18	29.48	0.505	187.32	16.43
S^4DL [31]	0.914	77.12	15.91	1.086	157.93	28.06	1.112	116.75	14.13
Diff-MEA (Ours)	0.536	45.91	14.35	0.535	46.50	33.76	0.535	46.78	16.96

2) Balancing Coefficient λ_a: The balancing coefficient λ_a was tested with five values: $\{100.0, 10.0, 1.0, 0.1, 0.01\}$. As shown in Fig. 11(b), the Houston and SH2HZ tasks achieved optimal performance at $\lambda_a = 1.0$, while the Indiana task performed best at $\lambda_a = 10.0$. This suggests that the samples in the Houston and SH2HZ datasets exhibit compact intraclass distributions and clear interclass separation, enabling robust discriminative signals with smaller semantic constraint weights. On the contrary, the Indiana dataset, which exhibits weaker semantic signals, necessitates a larger λ_a to preserve class-specific information throughout the generation process.

3) Number of 3-D Convolutional Layers in DDPM: To ascertain the optimal network depth for the diffusion model, the number of 3-D convolutional layers is varied within $\{1, 2, 3, 4, 5\}$. As depicted in Fig. 11(c), an architecture with three layers consistently yields the highest OA across all three datasets. The steep performance gradients observed indicate that the model is highly sensitive to this architectural parameter. A shallow configuration restricts the receptive field, thereby impeding the extraction of long-range spatial–spectral dependencies and leading to residual artifacts during the reverse diffusion phase. Conversely, excessive network depth induces severe spatial–spectral feature compression. The aggressive, repeated 3-D down-sampling in deeper networks irreversibly truncates high-frequency details, ultimately resulting in oversmoothed reconstructions.

4) Feature Dimension C: Finally, the dimension C of the latent feature space is assessed across five capacities: $\{32, 48, 64, 80, 96\}$, utilizing the predetermined optimal DDPM configuration of three layers. The trajectories in Fig. 11(d) confirm that all three datasets converge to a performance maximum at $C = 64$. Furthermore, the OA exhibits a rapid initial ascent as C increases from its minimum tested values. This phenomenon implies that an overly constrained feature space introduces a severe information bottleneck, fundamentally restricting the model's representational capacity and precluding the retention of fine-grained spectral characteristics.

G. Analysis of the Computational Complexity

We compare the computational complexity (parameters, FLOPs, and inference time) of the Diff-MEA and other methods using an NVIDIA RTX 4090 GPU and an Intel Xeon Silver 4210R CPU. As shown in Table VIII, Diff-MEA maintains approximately 0.53 M parameters and 46 M FLOPs across the three datasets, effectively avoiding a heavy model structure. This significantly reduces computational overhead compared to parameter-heavy (e.g., TSTnet) or FLOP-intensive (e.g., S2PST) models. Moreover, unlike mainstream UDA methods that necessitate loading all TD samples for simultaneous cross-domain alignment, Phase III of the Diff-MEA relies exclusively on source domain training samples. This data decoupling drastically minimizes the memory footprint and enhances batch processing efficiency. Furthermore, despite intensive generative training, the Diff-MEA achieves highly competitive inference speeds, requiring only 14.35 and 16.96 s on the Houston and Indiana datasets, respectively, yielding a favorable performance-efficiency tradeoff.

V. CONCLUSION

In this article, we introduced the Diff-MEA, a generative framework designed to address the fundamental challenges of feature topology collapse and semantic drift in cross-domain HSI classification. Departing from conventional single-step hard alignment strategies that are vulnerable to severe distribution discrepancies, the proposed approach reformulates the UDA as a continuous probabilistic manifold evolution process. This formulation is realized through the SSBCE mechanism, which tightly integrates the generative manifold exploration capability of the DDPM with the discriminative BA of the MTDA. By anchoring stable spatial–spectral representations extracted from a pretrained Mamba-based encoder, the proposed alternating cyclic optimization progressively decomposes large-scale domain discrepancies into a sequence of manageable microlevel transitions, enabling the construction of a smooth transitional domain while effectively mitigating semantic degradation. Extensive experiments conducted on three challenging cross-scene benchmarks, namely Houston, SH2HZ, and Indiana, validate the effectiveness of the proposed framework. Under unified experimental protocols, Diff-MEA consistently achieves superior performance across all major evaluation metrics, demonstrating strong capability in preserving class-discriminative structures and accurately delineating semantic boundaries even under pronounced geometric and distributional shifts. Future work will focus on integrating

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the current multistage optimization pipeline into a unified end-to-end training framework, aiming to reduce cross-phase transition overhead while maintaining the core coevolutionary optimization mechanism, thereby facilitating more efficient and scalable domain adaptation.

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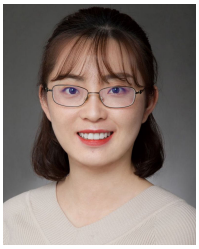
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